

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay.
- Best TSPLIB holdout gap: tsplib_failure_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_145329_n
- started_at_local: 2026-05-13 14:53:29
- finished_at_local: 2026-05-13 15:03:27
- duration_hhmm: 00:10
- duration_seconds: 597.297
- seed_offset: 13000
- replicate_label: n
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.096215	0.0	0.061228	0.900415	0.76	-0.001811
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.853264	0.56	-0.029188
tsplib_failure_replay	failure	score_only	False	0.077078	0.0	0.04905	0.853725	0.76	0.084983
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.096215, synthetic 0.0, combined 0.061228.

- Accepted-epoch count 2, mean accepted code novelty 0.900415, and final complexity 0.76.
- Adaptation efficiency -0.001811 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.096215 across 7 instances; family means: ch=0.077801, kroD=0.069597, pcb=0.113396, pr=0.188926, rd=0.097092, st=0.048889.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19928, "family": "lin", "name": "lin105", "optimality_gap": 0.38591}, {"best_known_cost": 629, "cost": 809, "family": "eil", "name": "eil101", "optimality_gap": 0.286169}, {"best_known_cost": 2579, "cost": 3296, "family": "a", "name": "a280", "optimality_gap": 0.278015}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.853264, and final complexity 0.56.
- Adaptation efficiency -0.029188 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.077078, synthetic 0.0, combined 0.04905.
- Accepted-epoch count 2, mean accepted code novelty 0.853725, and final complexity 0.76.
- Adaptation efficiency 0.084983 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.077078 across 7 instances; family means: ch=0.036917, kroD=0.03602, pcb=0.201505, pr=0.105132, rd=0.093426, st=0.02963.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3413, "family": "a", "name": "a280", "optimality_gap": 0.323381}, {"best_known_cost": 14379, "cost": 18619, "family": "lin", "name": "lin105", "optimality_gap": 0.294874}, {"best_known_cost": 7542, "cost": 8754, "family": "berlin", "name": "berlin52", "optimality_gap": 0.1607}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Results

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (Mean)	Complexity	Notes on Transfer and Novelty
tsplib_failure_replay	Failure Replay	**0.0771**	0.0	**0.0491**	0.854	0.76	Best transfer and TSPLIB gaps; zero synthetic gap; moderate code novelty
tsplib_no_replay	No Replay	0.0962	**0.0**	0.0612	**0.900**	0.76	Best synthetic gap (zero), slightly worse transfer and TSPLIB gaps; highest code novelty
tsplib_random_replay	Random Replay	0.2134	0.0649	0.1594	0.853	0.56	Poorer transfer and TSPLIB gaps; some synthetic gap; moderate code novelty
tsplib_failure_replay_compression	Failure Replay + Compression	0.2134	0.0649	0.1594	0.0	0.56	Matches random replay in gaps; zero code novelty; compression pressure evident

Interpretation

1. **Key Transfer Metrics (Held-out TSPLIB and Synthetic Holdout):**
 - The **tsplib_failure_replay** condition achieves the best transfer performance, evidenced by the lowest held-out TSPLIB gap (0.077) and transfer gap (0.049), with zero synthetic holdout gap.
 - **tsplib_no_replay** also achieves zero synthetic holdout gap but slightly higher transfer (0.061) and TSPLIB gap (0.096), indicating good but less strong transfer than failure replay.
 - Both **tsplib_random_replay** and **tsplib_failure_replay_compression** show substantially poorer transfer performance (TSPLIB gap ~0.213, transfer gap ~0.159), indicating worse generalization.
2. **Code Novelty vs. Transfer:**
 - Despite the **tsplib_failure_replay** condition having lower code novelty (mean 0.854) than no replay (mean 0.900), transfer performance improves significantly. This indicates that **improved transfer is not due to increased code novelty but better replay strategy (failure replay).**
 - The **failure replay compression** condition has zero code novelty and poorer transfer, indicating that compression pressure and novelty gating limit algorithmic innovation and degrade transfer.

- Random replay has moderate code novelty (~0.853) but performs worse on transfer gaps, suggesting that random replay is less effective despite some novelty.

3. **Effect of Replay Mode:**

- **Failure replay** (no compression) clearly outperforms no replay and random replay in transfer and holds zero synthetic gap.
- **No replay** yields excellent synthetic generalization but inferior TSPLIB transfer.
- **Random replay** results in poorer transfer and synthetic generalization.
- **Failure replay with compression** degrades both transfer and novelty, negating the benefits of failure replay.

4. **Complexity and Behavior:**

- Higher complexity (0.76) was observed in no replay and failure replay conditions, associated with better transfer.
- Lower complexity (0.56) in random replay and failure replay compression correlates with worse transfer.

Conclusions

- **Failure replay without compression yields the best transfer performance on held-out TSPLIB instances and synthetic benchmarks, outperforming no replay and other replay methods.**
- **Improved transfer occurs despite a slight reduction in code novelty compared to no replay, showing that replay strategy, not lexical novelty alone, drives this improvement.**
- **Compression-aware failure replay severely limits code novelty and transfer performance, indicating a detrimental trade-off.**
- **Random replay is less effective for transfer despite moderate code novelty.**

These results highlight the importance of carefully distinguishing replay types: **failure replay optimizes for transfer**, while compression pressure unduly restricts learned solutions and harms performance. Optimality gaps and transfer metrics provide the primary evidence; narrative speculation should refrain from equating code novelty with algorithmic novelty without this support.